Full-Cycle Secular Scaling for an Open Walsh Gate

Route-A structural certificate C158

Abstract

At the natural full-cycle time of a frozen open Walsh shift, the propagator is an exact tensor power. We obtain its complete secular factorization and a binomial law for surviving spectral log moduli. The law concentrates at $-\log(3)/4$ and has Gaussian fluctuations. These are source-side scattering results, not a self-adjoint or target construction.

1 Full-cycle factorization

Let F_3 be the normalized three-point DFT, $P = \operatorname{diag}(1, 0, 1)$, and $A = F_3^* P$. On $(\mathbb{C}^3)^{\otimes k}$ define

$$B_k(v_0, \dots, v_{k-1}) = (v_1, \dots, v_{k-1}, Av_0).$$

One application is one tick. After exactly k ticks, every original factor has passed through A once and returned to its position, hence

$$C_k := B_k^k = A^{\otimes k}. \tag{1}$$

The one-site characteristic polynomial is

$$\chi_A(\lambda) = \lambda(\lambda^2 - \tau\lambda + q), \quad \tau = \frac{\sqrt{3}}{6} - \frac{i}{2}, \quad q = -\frac{1}{2} - \frac{\sqrt{3}}{6}i.$$

Its discriminant is $11/6 + \sqrt{3}i/2$, so the roots $0, \lambda_+, \lambda_-$ are distinct. Label the nonzero roots by $|\lambda_+| > |\lambda_-|$. Tensoring a diagonalization and counting binary words gives

$$E_k(z) := \det(I - zC_k) = \prod_{j=0}^{k} (1 - z\lambda_+^j \lambda_-^{k-j})^{\binom{k}{j}}. \tag{2}$$

Thus $\deg E_k = 2^k$, and the zero generalized eigenspace has dimension $3^k - 2^k$.

2 Surviving log-modulus scaling

Choose a nonzero eigenvalue ρ with algebraic-multiplicity weight and fix the convention

$$X_k = k^{-1} \log |\rho|.$$

We do not replace this by an inverse secular-zero radius or by a phase. Formula (2) makes

$$X_k = \frac{J_k a + (k - J_k)b}{k}, \qquad J_k \sim \operatorname{Bin}(k, 1/2),$$

where $a = \log |\lambda_+|$ and $b = \log |\lambda_-|$. Since $|\lambda_+\lambda_-| = |q| = 3^{-1/2}$,

$$\mathbb{E}X_k = \mu = -\frac{\log 3}{4}, \qquad \operatorname{Var}(X_k) = \frac{\sigma^2}{k}, \quad \sigma^2 = \frac{1}{4}\log^2 \frac{|\lambda_+|}{|\lambda_-|}. \tag{3}$$

The Bernoulli concentration estimate gives, for $\epsilon > 0$,

$$\Pr(|X_k - \mu| \geq \epsilon) \leq 2\exp\left(-\frac{k\epsilon^2}{2\sigma^2}\right). \tag{4}$$

Therefore the empirical measures converge weakly to δ_μ. For the fluctuation law, let Y_ℓ be independent fair signs. Then $\sqrt{k}(X_k - \mu) = (a - b)(2\sqrt{k})^{-1} \sum_{\ell=1}^k Y_\ell$, whose characteristic function is $\cos(t(a - b)/(2\sqrt{k}))^k \to e^{-\sigma^2 t^2/2}$. Hence

$$\sqrt{k}(X_k - \mu) \Longrightarrow N(0, \sigma^2). \tag{5}$$

The moduli are genuinely unequal. Writing $p_\pm = |\lambda_\pm|^2$, direct algebra gives

$$p_+ + p_- = \frac{1 + \sqrt{37}}{6}, \quad p_+ p_- = \frac{1}{3}, \quad (p_+ - p_-)^2 = \frac{\sqrt{37} - 5}{18} > 0.$$

3 Exact receipts and boundary

Exact $\mathbb{Q}(\sqrt{3}, i)$ Newton coefficients are frozen through $k = 5$, direct Kronecker receipts through $k = 3$, and binomial moments through $k = 24$. The independent checker, SymPy path, replay, and hostile mutations reconstruct these finite sentinels; the all-k theorem comes from (1)–(5).

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